

Ishan Trivedi

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PROFILE

Second-year Computer Science undergraduate focused on applied AI engineering, LLM fine-tuning, RAG systems, local model serving, agentic workflows, and backend systems. Built a QLoRA fine-tuned Qwen2.5 log-analysis adapter, an incident-investigation RAG backend, a local LLM server, and API-backed automation systems. Comfortable taking AI ideas from dataset design to runnable training, inference, retrieval, evaluation, API integration, documentation, and deployment-oriented engineering.

PROJECTS

- IncidentOps - RAG Backend for Incident Investigation** [GitHub] 2026 – Present
- Building a FastAPI backend for engineering incident investigation using normalized document ingestion, hybrid retrieval, and citation-backed answers.
 - Implemented project-scoped ingestion, chunking, metadata extraction, pgvector search, PostgreSQL full-text search, evidence packing, and deterministic investigation flows.
 - Added workflow run persistence, approval gates, eval endpoints, metrics, redaction, prompt-injection checks, and smoke-testable API flows.
- LogSage - QLoRA Fine-Tuned Log Incident Analyzer** [GitHub] | [Hugging Face] 2026
- Fine-tuned a Qwen2.5-7B-Instruct adapter using QLoRA for structured log triage with JSON outputs: issue, root cause, severity, fix, and confidence.
 - Built and validated a 1,116-row supervised instruction dataset with ChatML-style prompts, schema normalization, and severity/confidence consistency.
 - Ran training on AWS EC2 GPU infrastructure, tracked train/eval losses, TensorBoard metrics, checkpoints, and published the adapter with reproducible Colab/local inference.
 - Implemented Python inference utilities for adapter loading, generation, JSON extraction, and schema validation instead of relying on raw model text.
- llm_server - Local LLM Server and Tactical AI Interface** [GitHub] 2026
- Built a local-first AI assistant stack with a Python backend and TypeScript frontend for running private LLM interactions on personal hardware.
 - Designed the backend to manage local model execution, response generation, memory/state handling, and API communication without relying on hosted chatbot subscriptions.
 - Developed a tactical terminal-style chat interface for visualizing local AI interactions and providing a polished user experience.
- Acali - Agentic Telegram Assistant for Calendar Automation** [GitHub] 2025
- Developed a Python/FastAPI backend that connects Telegram interaction flows with Google Calendar automation.
 - Integrated Google OAuth 2.0, persistent state, secure token handling, and LLM-based intent parsing for scheduling workflows.
 - Designed agent-style flows for event creation, conflict detection, contextual memory, and webhook-based local testing.

TECHNICAL SKILLS

Programming: Python, Java, TypeScript basics

AI / ML: PyTorch, Hugging Face Transformers, PEFT/LoRA/QLoRA, CNNs, supervised fine-tuning, model evaluation basics

LLM / RAG / Agents: LangGraph, Pydantic structured outputs, tool-calling workflows, RAG pipelines, hybrid retrieval, prompt design, JSON schema validation

AI-Assisted Engineering: Proficient with ChatGPT, Codex, Claude, and equivalent AI coding/research assistants.

Backend / Infra: FastAPI, OAuth 2.0, PostgreSQL, SQL basics, pgvector, Docker basics, AWS EC2, Git/GitHub

Data: NumPy, pandas, scikit-learn, JSON/JSONL, Jupyter/Colab

DSA: Proficient in arrays, strings, hashing, recursion, sorting/searching, two pointers, and basic complexity analysis

EDUCATION

- Sir M. Visvesvaraya Institute of Technology, Bengaluru** 2024 – Present
- B.E. in Computer Science & Engineering, Visvesvaraya Technological University CGPA: 8.5/10
 - Relevant Coursework: Data Structures, Linear Algebra, Discrete Mathematics, Object-Oriented Programming
- Pandya Memorial High School, Rajasthan** 2022 – 2024
- Higher Secondary Education, Mathematics-Science Stream 91.5%

CERTIFICATIONS & APPLIED LEARNING

Hugging Face LLM Course - Transformers, Datasets, Tokenizers, Accelerate, Hub workflows, fine-tuning, and model sharing.

Hugging Face AI Agents Course - agents, tool use, smolagents, LlamaIndex, LangGraph, agentic RAG, observability, and evaluation.

DeepStation Community - participant in AI systems-focused learning community.

LANGUAGES

English (Professional working proficiency), Hindi (Native)